

FOC ESC — PCB Layout & Stackup Guide

STM32G473 · DRV8328 · inline shunts + INA240 · AS5047P · Ø60 mm 4-layer board · 24 V / 10 A pk / 5–8 A cont · JLCPCB fab

1. Final architecture & review verdict

Placement verdict — ACCEPTED with refinements. "Everything on top, encoder alone on bottom" is the proven layout for this class of board (moteus r4 is Ø46 mm and does exactly this). It buys you: single-side assembly (cheaper JLC SMT run, one reflow pass), an almost-empty bottom that becomes a clean magnetic/electrical quiet zone, and free heat-spreading copper on the bottom. Cost: the top layer is crowded — placement discipline decides whether this board works.

Stackup verdict — ACCEPTED with one change. The proposed stack (top: everything / GND / power / bottom: encoder) is correct in structure. Change: L3 should not be a wall-to-wall power plane. Make L3 **GND fill with power islands** (24 V island under the bridge, 5 V/3.3 V islands where needed). Reason in §3.

Copper verdict — 2 oz outer / 0.5 oz inner. Your trace calculation is correct but answers the wrong question. Detail in §3.3.

Block	Layer	Zone (see Fig. 2)	Driving constraint
6× MOSFETs, 3× shunts, phase pads	Top	Power arc, north edge	Hot loop area, heat spreading, short phase exits
DRV8328 + bootstrap/ GVDD caps	Top	Center-north, inside the bridge arc	Equal short gate traces to all 6 FETs
Bulk electrolytic + per-phase MLCCs	Top	At the bridge, on the PVDD entry side	Minimize switching loop; MLCCs closest, bulk behind
3× INA240	Top	Directly south of shunts	Kelvin sense trace length; amplified outputs travel, raw shunt signals must not
5 V buck (IC + inductor)	Top	East edge, near power entry	Own hot loop; keep SW node away from INA240 inputs & encoder axis
3.3 V LDO	Top	Between buck and MCU	Input from 5 V, output rail feeds analog/digital star
STM32G473 (LQFP64) + crystal	Top	Center-south quiet zone	Short ADC runs from INA240s; short SPI drop to encoder vias; crystal <10 mm from MCU
SN65HVD230 + CAN connector	Top	West edge	Minimal CANH/CANL stub; connector at edge
USB-C + USBLC6 + CC pulldowns	Top	South edge	TVS at connector; D± pair to MCU without crossing power zone
AS5047P + 2 decoupling caps	Bottom	Dead center on rotation axis	Magnet alignment; keep-out ring; nothing else on bottom

2. Placement plan

TOP — component side (faces away from motor) BOTTOM — encoder side (faces motor / magnet)

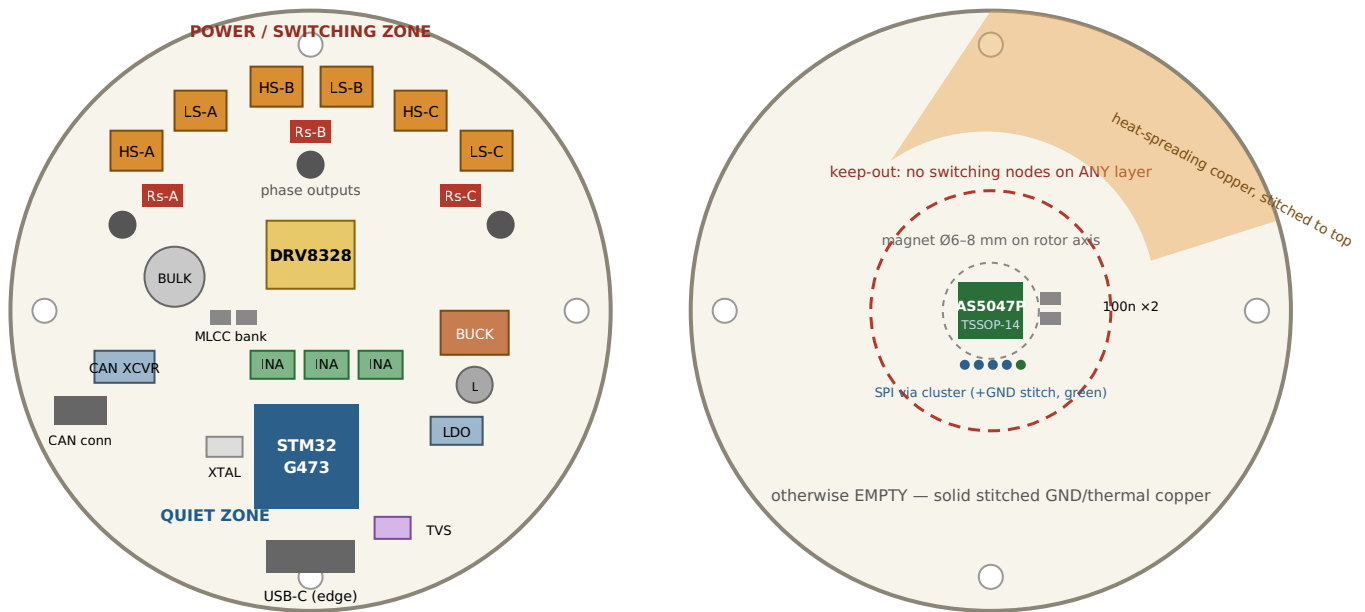


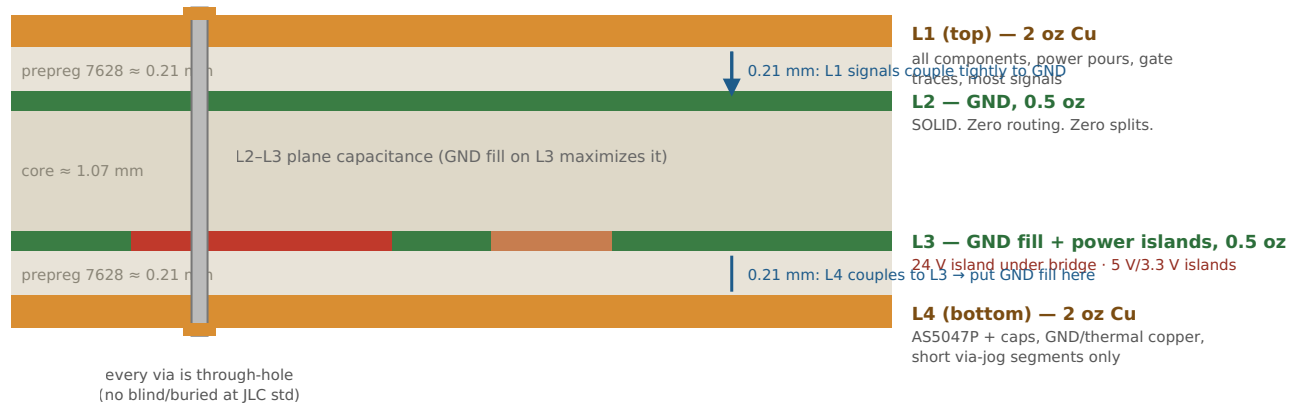
Fig. 2 — Placement map, Ø60 mm. Power arc north, quiet zone south, encoder alone on bottom center. Mounting holes shown for motor standoffs — fix these and the connector positions *first*, they are mechanical constraints.

2.1 Placement rules & reasoning

- **Partition, don't moat.** One contiguous ground everywhere; separation is *spatial* — switching components clustered in one region, sensitive components in another, so their return currents occupy different areas of the same plane. TI's motor-driver layout app note (SLVA959) is explicit that this is partitioning of *placement*, not split grounds.
- **Order of placement:** ① mounting holes + connectors + encoder axis (mechanical, immovable) → ② FET bridge + shunts + phase pads as one tight arc → ③ DRV8328 centered on the bridge → ④ decoupling at the bridge → ⑤ INA240s against the shunts → ⑥ MCU so that ADC inputs face the INA240 outputs and SPI pins face the center → ⑦ everything else.
- **The buck is a second noise source.** Treat it like a fourth half-bridge: its input hot loop and SW node get the same care, and it must not sit between the shunts and the INA240s, nor within the encoder keep-out cylinder.
- **Crystal goes on the MCU's side facing away from the power arc.** It is not magnetically sensitive, but its traces are high-impedance nodes — capacitive pickup from dv/dt nodes (phase, SW) is the risk.
- **USB-C and CAN connectors on the south/west edge** — physically opposite the phase wiring so motor leads and comm cables leave the board on different sides.

Search: moteus r4 board layout · SLVA959 motor driver layout best practices ·
PCB floorplanning power analog partitioning

3. Stackup & copper weight



JLPCB reference: standard 4-layer stack (7628 prepreg / 1.065 mm core). Select the 2 oz-outer variant in the stackup picker; inner layers stay 0.5 oz. Verify exact dielectric numbers in the JLPCB impedance/stackup table for your order.

Fig. 1 — Stackup. Physical orientation: “top” faces away from the motor, “bottom” (encoder) faces the magnet.

3.1 Why signal-GND spacing is the whole game

Why: every signal trace is one half of a transmission line; the return current flows in the copper *directly underneath it*, concentrated within roughly $\pm 3\times$ the dielectric height. With 0.21 mm to the plane, the loop area of every top-layer trace is tiny → low inductance, low emission, low pickup. This is why sig-GND--GND-sig style stacks are recommended: **both** routing layers get a close reference. Your stack achieves the same thing as long as L3 under the bottom routing regions is ground fill — which is exactly why L3 must not be a solid 24 V plane.

3.2 Why L3 = GND fill with power islands (not a full power plane)

- Anything routed on L4 (SPI jog, USB jog, encoder caps) would otherwise reference the 24 V plane. Return current would have to displace through the L2-L3 plane capacitance to get back to GND — a larger, lossier, noisier loop.
- A 24 V island only needs to exist under the bridge/bulk-cap region. 0.5 oz plane copper has $\approx 1 \text{ m}\Omega/\text{square}$; even a modest island carries the DC feed with millivolt drops. The *switching* (AC) current never needs L3 at all — it circulates in the top-layer hot loop (§5).
- GND fill on L3 doubles as the second reference plane and adds interplane capacitance across the thick core.

3.3 Copper weight — critique of your calculation

Your number ($\approx 11 \text{ mm}$ of 1 oz for 10 A at 5 °C rise) is a correct IPC-2221-style *ampacity* result, and it proves ampacity is not the constraint — on a Ø60 mm board you will use pours, and pours are far wider than 11 mm. The actual reasons to pay for 2 oz:

- The copper is your only heatsink.** $\sim 0.3\text{--}0.5 \text{ W}$ per conducting FET plus buck and shunt losses $\approx 2\text{--}3 \text{ W}$ total must spread laterally from a few mm^2 of PDFN pads into the pour before convection can remove it. Lateral thermal resistance of copper scales inversely with thickness: 2 oz halves the spreading resistance. This is the difference between a warm board and a hot spot.
- Necking.** Pour cross-section collapses where it squeezes between pads, vias, and clearance holes. 2 oz keeps those pinch points cool. Trace calculators model a uniform trace and miss this entirely.
- Precedent:** moteus, ODrive and VESC-class boards at similar or higher current all use 2 oz outer.

Cost of 2 oz: JLCPCB minimum trace/space grows to about 0.20 mm on 2 oz outer layers (vs 0.09 mm at 1 oz). Check your finest-pitch footprint: LQFP64 at 0.5 mm pitch leaves ≈ 0.2 mm between pads — at the limit but routable (don't run traces between LQFP pads; escape straight out). TSSOP (INA240, AS5047P) and the PDFN FETs are fine. If DRC shows violations you cannot resolve, 1 oz + maximum pour area + dense stitching to the near-empty bottom copper is an acceptable fallback at 5 A continuous — derate continuous current expectations, not peak.

Search: IPC-2152 current carrying capacity vs IPC-2221 ·

copper pour thermal spreading resistance · JLCPCB 4 layer stackup 7628 impedance ·

JLCPCB 2oz outer copper minimum trace width · plane capacitance power ground pair

4. Grounding, vias & return paths — the rules

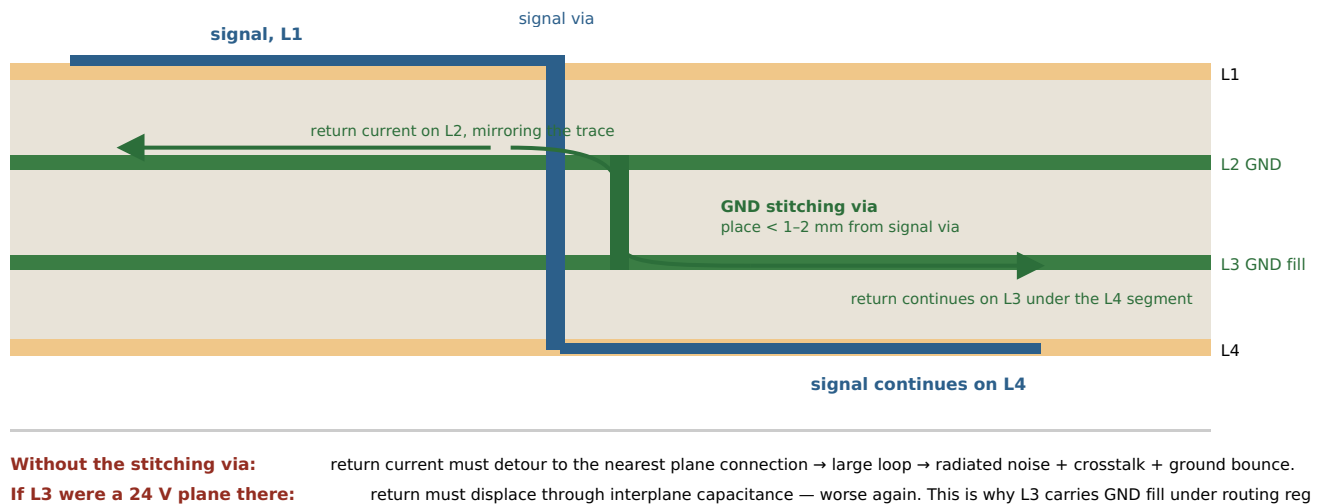


Fig. 3 — What happens at a layer transition. The signal changes layers instantly; its return current can only follow if you give it a ground path at the same spot.

4.1 Rules, each with its reason

R1 — L2 is sacred. No traces, no splits, no via rows that chain antipads into slots.

Why: a slot in the reference plane forces every crossing trace's return current to flow around the slot end — turning a 0.2 mm loop into a multi-cm loop. One routed trace on L2 can undo the entire benefit of the 4-layer board. Space plane vias ≥ 0.5 mm apart so copper webs remain between antipads.

R2 — Signals change layers only between L1 and L4, never "onto" L2/L3. Your USB question, answered: the D− jog under D+ goes on the **opposite signal layer (L4)**, short, with GND fill around it, plus one GND stitching via beside the via pair.

Why: plane layers are references, not routing resources. Routing on them cuts them (see R1). Since JLC standard service is through-vias only, any via already reaches L4 for free — use it.

R3 — Every signal layer-change gets a GND via within 1-2 mm. For high-speed or clock-bearing nets (SPI SCK, USB), make this non-negotiable; for slow GPIO it's good hygiene.

Why: Fig. 3. The stitching via is the bridge that lets the return current change planes with the signal.

R4 — Ground planes/pours are one net, stitched everywhere. L2, L3-fill, plus top and bottom GND pours: perimeter ring of vias at 3–5 mm pitch, dense field (1–1.5 mm pitch) under and around the FETs, a stitch at every GND pad of every decoupling cap.

Why: stitching serves three jobs at once — return-path continuity, low-impedance ground (many vias in parallel), and thermal conduction into inner/bottom copper. Under the FETs the vias are doing all three.

R5 — High current lives in pours on L1, not traces, and not on inner layers. PVDD pour → bridge → shunts → phase pads: all top copper. The L3 24 V island is DC distribution/backup only.

Why: inner copper is 0.5 oz (4× the sheet resistance of your 2 oz outer) and thermally landlocked. Pours also have far lower inductance than any trace. Use direct connect (no thermal relief) on all power-net pads except where hand-rework matters.

R6 — Via budget for current: ~1 A per 0.3 mm via, conservatively. Where phase current must reach bottom-side thermal copper, use ≥ 8 –10 vias per node.

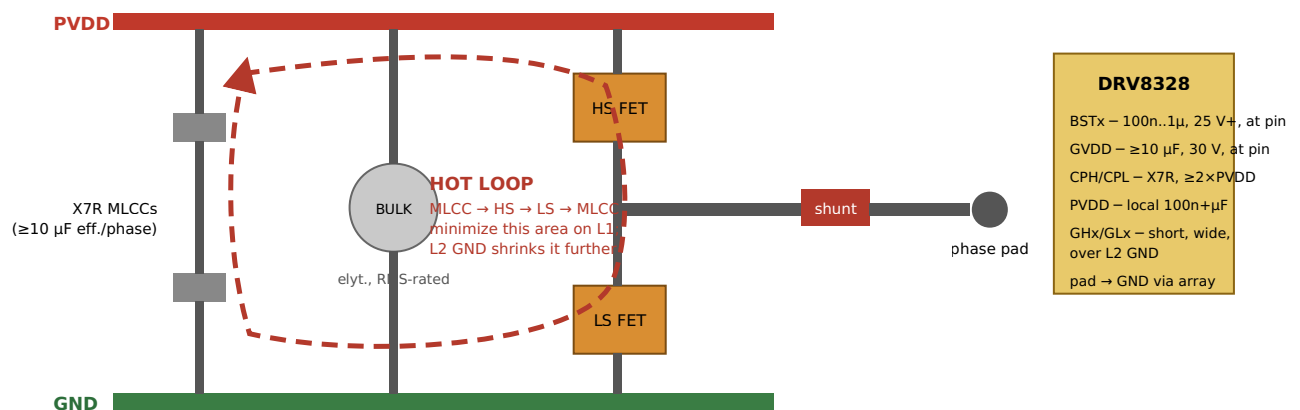
Why: a 0.3 mm drill via with standard plating is roughly 1–2 mΩ; paralleling is cheap. The number matters more for heat than for resistance.

4.2 Which layer does each net actually use?

Net / group	Layer plan	Reason
Gate drives (6×), bootstrap, SHx sense	L1 only, no vias	High di/dt into gate; via inductance causes ringing. DRV placement makes this possible.
PVDD, phase, GND power path	L1 pours (+ stitched L4 thermal copper)	R5, R6
Shunt sense pairs → INA240	L1 only, as tight diff pairs	μV-level signals; vias add asymmetry and pickup. Placement makes vias unnecessary.
INA240 outputs → ADC pins	L1; L4 jog allowed if unavoidable	Amplified, low-impedance — robust. Still keep away from SW/phase nodes.
SPI to encoder	L1 from MCU → via cluster near center → L4 short run to AS5047P	Must cross the board; one clean transition with a GND stitch (Fig. 2, via cluster), L3 has GND fill below the L4 run.
USB D+/D–	L1; the one crossing jog on L4 (\$7)	R2
CANH/CANL	L1, short, side-by-side	Differential pair; keep stub from transceiver to connector minimal.
24 V feed, 5 V, 3.3 V distribution	L3 islands + L1 where convenient	Low current; islands keep L4 regions referenced to GND fill.

Search: [return current path reference plane change via](#) · [ground plane slot antipad EMI](#) · [via stitching guidelines high speed](#) · [Rick Hartley how return current flows](#) (conference talk — the single best hour on this topic) · [thermal via array PDFN current capacity per via](#)

5. Power stage — bridge + DRV8328



Repeat ×3. The dashed loop carries the full switched current with ~ns edges — its area sets your EMI and phase-node ringing. Place the MLCCs so this loop closes in millimetres on L1; the bulk cap sits just behind them and only handles low-frequency ripple.

Fig. 4 — One half-bridge with its decoupling and the switching (“hot”) loop. Note the shunt is in the **phase output** (inline sensing), outside the hot loop — one of the benefits of in-line shunts over low-side.

5.1 DRV8328 datasheet layout requirements (extracted)

Item	Requirement	Placement note
Bootstrap caps (BSTx-SHx)	Ceramic, rated ≥ 25 V (bootstrap rail is 12–13 V above SHx; TI: rate caps $\geq 2 \times$ working voltage for DC-bias margin)	Directly at BSTx/SHx pins; the SHx side connects to the phase node copper with the shortest possible link
GVDD cap	X5R/X7R, $\geq 10 \mu\text{F}$, 30 V-rated; value $> 10 \times C_{\text{BST}}$	At GVDD–GND pins; this cap sources every gate pulse — treat like a decoupler, not a bulk cap
Charge-pump cap (CPH–CPL)	X5R/X7R, rated $\geq 2 \times \text{PVDD}$	Directly between CPH and CPL pins
PVDD decoupling	Local ceramic + bulk on the driver's PVDD pin	At pin, via to 24 V
Gate traces GHx/GLx (+SHx returns)	Short, wide, minimal loop	This is why DRV8328 sits centered inside the bridge arc; route each gate with its return (SHx for high side, GND for low side) side-by-side on L1
Thermal pad	Tie to GND, via array to planes	TI thermal app-note figure: ~0.5 mm pad / 0.2–0.3 mm drill vias under the pad; on JLC use 0.3 mm drill, accept minor wicking or pay for filled+capped vias
Optional power-stage tuning parts	Gate-source pulldown Rs, phase RC snubbers, PVDD TVS	Reserve footprints now (DNP); the datasheet's "system considerations" section lists these as debug/tuning aids — you cannot add them after routing a Ø60 mm board

5.2 MOSFET bridge rules

- **Arrange each half-bridge as one vertical strip:** PVDD pour → HS drain pad; HS source → phase copper → LS drain; LS source → GND pour. Phase copper connects onward to shunt → phase pad. Minimal jumps, zero vias in the current path on L1.
- **Per-phase MLCC bank between PVDD and GND at each half-bridge** — target $\geq 10 \mu\text{F}$ effective at 24 V bias (means ~2–3× nominal after DC-bias derating; e.g. $3 \times 10 \mu\text{F}/50 \text{ V}$ X7R 1210 per phase). Place them to physically close the Fig. 4 loop.

- **Bulk electrolytic:** sized by RMS ripple-current rating at your PWM frequency (with frequency derating), not by μF . Position at the PVDD entry into the bridge region.
- **Phase node copper: as much as needed for cooling, as little as possible for EMI.** It swings 0–24 V at ns edges; every cm^2 is a transmit antenna and a capacitive aggressor. Do not pour it large, do not run it near sense lines, keep it outside the encoder keep-out cylinder on all layers.
- **Thermal strategy per FET:** HS drain pad = PVDD (quiet) — spread this pour generously and stitch to bottom copper. LS drain pad = phase node — give it a modest local pour plus a small stitched bottom patch, placed outside the encoder keep-out.
- **Hardware OCP path:** INA240 outputs → G473 internal comparators (DAC threshold) → TIM1 BKIN. Keep the comparator input traces on L1 in the quiet zone. Verify DRV8328 variant fault pins (DT / OCP / DRVOFF depend on A/B/C/D variant) are wired per your chosen variant.

Search: half bridge hot loop layout minimization · MLCC DC bias derating X7R ·
 electrolytic capacitor ripple current frequency derating ·
 gate driver bootstrap capacitor layout · phase node ringing snubber design ·
 SLVA959B thermal via recommendation

6. Current sensing — inline shunts + INA240

✗ no vias in the sense path ✗ no current-carrying copper between tap points ✗ don't tap the pour — tap the pad

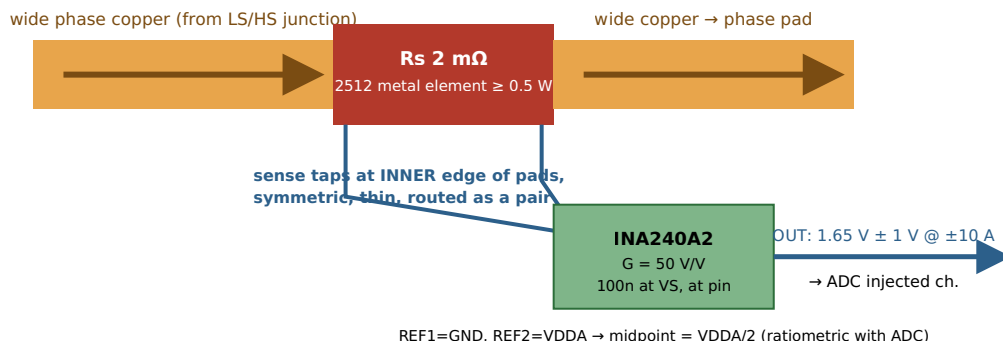


Fig. 5 — Kelvin connection. The measurement sees only the shunt element, never your copper (Cu is ~ 3900 ppm/ $^{\circ}$ C — 80 $\times$ worse tempco than the shunt).

6.1 INA240 datasheet layout requirements (extracted)

Item	Requirement
Shunt connection	Kelvin / 4-wire, per datasheet §"Connection to the Current-Sense Resistor" — any shared current-carrying impedance appears as gain/offset error against a 2 mΩ element
Bypass	0.1 μ F at VS-GND, at the pin; VS = 3.3 V rail (VDDA) so output range matches ADC
REF pins	Output midpoint = $(V_{REF1} + V_{REF2})/2$. Use REF1=GND, REF2=VDDA → 1.65 V midpoint that tracks the ADC reference (ratiometric — offset drift of the rail cancels in the conversion). Firmware still calibrates the true zero code at startup with the bridge disabled.
Input filtering	None required — the part is designed for direct PWM-node connection (enhanced PWM rejection). If you add RC anyway, keep R small (< 10 Ω) and matched; mismatch converts common-mode transients to differential error.
Placement	As close to the shunt as the power arc allows; sense pair length matters less than symmetry and distance from dv/dt nodes

6.2 Scaling check (2 mΩ + A2 gain)

± 10 A $\times$ 2 mΩ $\times$ 50 V/V = ± 1.0 V around 1.65 V → 0.65–2.65 V at the ADC: inside the linear output range (swing limit $\approx$ VS–0.2 V) with headroom for overcurrent detection above 10 A. Dissipation 0.2 W peak / 0.13 W at 8 A — a 0.5 W-rated 2512 metal-element part runs warm but in spec; 1 W rating is cheap insurance. A1 (20 V/V) wastes 60 % of ADC range; A3 (100 V/V) clips before your OCP threshold. **A2 is correct.**

Search: kelvin connection shunt 2512 layout · INA240 enhanced PWM rejection inline phase sensing · ratiometric ADC measurement reference tracking · current shunt tempco copper trace error

7. USB-C + USBLC6-2

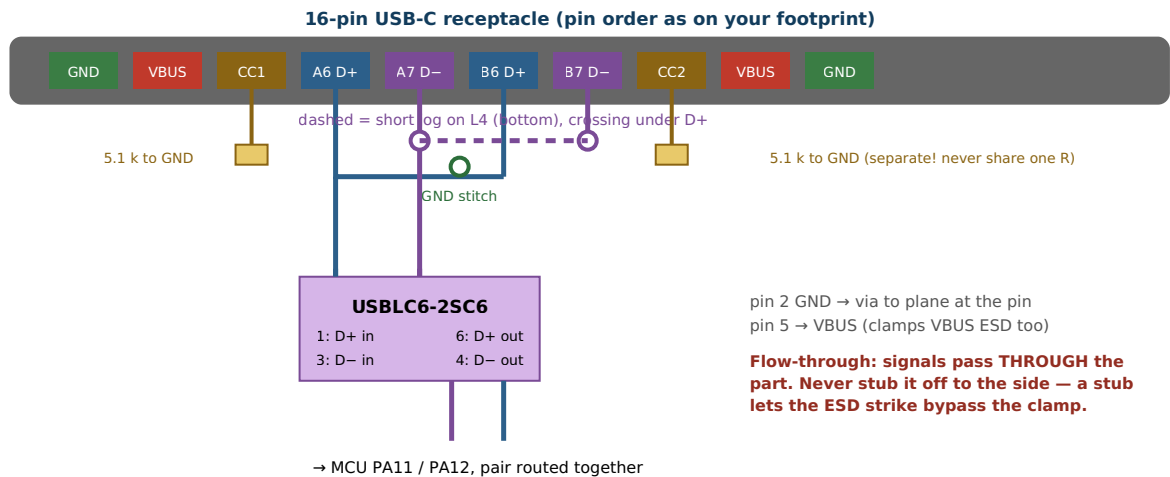


Fig. 6 — D± pair merging, the single L4 crossing jog, and flow-through TVS routing.

- **Yes, tie the pairs:** $A6+B6 = D+$, $A7+B7 = D-$. Mandatory, not optional — the host drives whichever pair the plug orientation lands on. Merge them *at the connector*, before the TVS, so both stubs stay short.
- **The crossing:** exactly one net needs a two-via jog. Route it on L4 (R2), a few mm long, over the L3 GND fill, one GND stitch via beside the pair. At full speed (12 Mbps, ~10 ns edges) the stub and impedance discontinuity are negligible — do not lose sleep over 90 Ω matching on this board.
- **CC1 and CC2 each get their own 5.1 k Ω to GND** (UFP role). Sharing one resistor breaks orientation detection and is the classic USB-C mistake.
- **VBUS:** goes only to detection/(optional) 5 V-input OR-ing — never to the 24 V domain. STM32G4's USB has integrated pull-up; no external 1.5 k Ω , no series resistors needed.
- **Shield/shell:** tie to GND (SELV board, chassis = GND); optionally via 1 M Ω || 1 nF footprint if you want the option to experiment.

Search: USB-C 16 pin UFP CC pulldown 5.1k · USBLC6-2 flow through layout ·
USB full speed routing requirements relaxed · ESD protection placement at connector why

8. Encoder zone — AS5047P

- **Alignment is the spec that bites.** Package center on the rotation axis; the die tolerates only a small displacement radius (~ 0.25 mm class) before angle error grows. Add a fab-drawing note + use the motor mounting holes as the datum. Diametric magnet (eval-kit reference: $\varnothing 6 \times 2.5$ mm NdFeB; datasheet reference: $\varnothing 8 \times 3$ mm N35H), airgap typically 0.5–3 mm — tune so the AGC register sits mid-range; MAGL/MAGH diagnostic flags tell you if the field is out of window. Read DIAAGC during bring-up.
- **3.3 V operation: VDD and VREG must be tied together**, decouple both pins (100 nF each) right at the part — these are the only other components on the bottom.
- **Keep-out cylinder, $\varnothing \sim 10$ –12 mm around the axis, through all layers:** no phase node, no SW node, no high-current pours. Solid GND on L2/L3 behind the sensor. The part rejects *homogeneous* stray fields; the near field of a phase trace 2 mm away is not homogeneous.
- **SPI:** dedicated bus (per your architecture), mode 1 (CPOL=0, CPHA=1), ≤ 10 MHz. Route: MCU (L1, quiet zone) → via cluster near board center → short L4 run to the part. One GND stitch via at the cluster. CS timing per datasheet (350 ns high time between frames).
- **No copper pour directly under the exposed sensor area is required or forbidden** — copper is non-magnetic; eddy-current shielding of a static/slow field is negligible at motor speeds. Ferrous materials (screws, standoffs) near the magnet are the actual risk — spec brass/aluminium standoffs near the center.

Search: AS5047P diametric magnet selection airgap AGC · magnetic encoder mounting tolerance angle error · ams AN000304 position sensor · eddy current effect magnetic encoder copper

9. Power supplies — 5 V buck + 3.3 V LDO

- **The buck's input hot loop (CIN → HS switch → sync FET/diode → CIN) is the noisiest few mm² after the bridge.** CIN ceramic directly across VIN-GND pins, same layer, no vias in the loop. SW node minimal copper: pad → inductor, nothing else.
- **Feedback trace:** from the output cap's positive terminal (sense the real output), thin, routed away from SW and the inductor; never under either.
- **Inductor orientation:** if shielded (use shielded), orientation is minor; still keep it off the encoder axis and > 10 mm from INA240 inputs.
- **24 V → buck input:** from the L3 24 V island or top pour, with local bulk ceramic — the buck must not share the bridge's decoupling.
- **LDO 5 V→3.3 V:** input/output caps per its datasheet (many LDOs have ESR/capacitance stability windows — check the specific part). Place output cap at the pin; distribute 3.3 V as an L3 island or top traces.
- **VDDA:** feed through a ferrite bead or $10\ \Omega$ from 3.3 V, with $1\ \mu\text{F} + 100\ \text{nF}$ at the MCU's VDDA pin. The INA240s' VS should come from the same VDDA branch for ratiometric consistency.

Search: buck converter layout input loop CIN placement · switch node copper area EMI tradeoff · feedback trace routing buck regulator · VDDA filtering ferrite bead STM32

10. MCU, crystal, CAN

- **STM32G473 decoupling:** 100 nF at every VDD/VSS pair (4 on LQFP64) + one $4.7\ \mu\text{F}$ bulk; VDDA per §9; VBAT to 3.3 V if unused. Caps on L1 beside the pins, each GND pad stitched.
- **Crystal (ST AN2867):** < 10 mm from OSC pins, symmetric short traces, load caps $C = 2 \cdot (C_L - C_{\text{stray}})$ with $C_{\text{stray}} \approx 3$ –5 pF, local GND pour around/under the crystal on L1 tied to L2 with 2–4 vias, no switching traces adjacent on any layer. Compute gain margin per AN2867 or copy a proven G4 reference design's crystal + caps.

- **NRST:** 100 nF to GND; **BOOT0:** 10 k pulldown; **SWD:** pads or Tag-Connect footprint — do not omit, USB DFU is not a debugger.
- **SN65HVD230:** 100 nF at VCC (3.3 V); RS pin: 10 k to GND for slope control (or GND for full speed); CANH/CANL as a side-by-side pair, transceiver adjacent to the connector so the stub off the bus is minimal. Termination: fit a split-termination footprint (60 Ω + 60 Ω with center 4.7 nF to GND), populate only if this node terminates the bus.
- **FDCAN routing to MCU:** ordinary digital — L1, quiet zone.

Search: AN2867 oscillator design guide STM32 · CAN bus split termination EMC · SN65HVD230 RS pin slope control · STM32 decoupling capacitor placement per pin

11. Study guide — one search per concept

Concept	Search phrase	What you should come away understanding
Return currents	Rick Hartley how return current flows in PCB	Return current mirrors the trace in the nearest plane; loop area is everything
Layer transitions	return path discontinuity via stitching	Why R3 exists; what happens at a plane change
Plane integrity	ground plane slot crossing trace EMI	Why one trace on L2 ruins the stackup
Hot loops	power stage commutation loop inductance	Which loop carries the di/dt and why MLCC placement sets ringing
Ampacity vs heat	IPC-2152 vs IPC-2221 derating	Why trace calculators overestimate needed width, and what they don't model (necking, pours)
Thermal design	PDFN thermal via array junction to ambient 4 layer	Copper spreading + via arrays are the heatsink
Kelvin sensing	4 wire kelvin shunt layout current sense amplifier	Why sense taps go on the pad, not the pour
DC bias derating	MLCC capacitance loss DC bias class II	Why "10 μ F" is 3–4 μ F at 24 V
Ripple current	electrolytic capacitor RMS ripple rating inverter	Why the bulk cap is chosen by A_{RMS} , not μ F
Magnetic encoders	on-axis magnetic encoder mounting tolerances	Airgap window, centering, ferrous keep-out
USB-C basics	USB-C legacy device CC resistors orientation	Why both $D\pm$ pairs and both CC pulldowns exist
ESD routing	TVS flow through routing ESD connector	Why the clamp must be first in line, in the path

12. Pre-fab checklist

Placement / mechanical

- ☐ Mounting holes match motor pattern; encoder package center on rotation axis (± 0.1 mm on the drawing)
- ☐ Magnet airgap achievable with chosen standoffs; non-ferrous standoffs near center
- ☐ USB-C / CAN connector accessibility checked against motor + wiring in 3D
- ☐ Bulk cap height clears the enclosure/motor mount

Stackup / fab

- ☐ JLC 4-layer, 2 oz outer / 0.5 oz inner selected; DRC rules updated to that stackup's min trace/space (≈ 0.2 mm outer)
- ☐ All vias through-hole; no via-in-pad except DRV8328 thermal pad (accept wicking or order filled/capped)
- ☐ L2 100 % solid — run a "no routes, no splits on L2" visual check as the final step before export
- ☐ L3 GND fill present under every L4 routing region and under the encoder

Power stage

- ☐ Hot loop per phase closes on L1 through local MLCCs; loop area visibly minimal
- ☐ Bootstrap, GVDD, CP caps at DRV8328 pins; ratings 25 V/30 V/2×PVDD respectively
- ☐ Gate traces L1-only, each with its return alongside
- ☐ DNP footprints reserved: gate pulldowns, phase snubbers, PVDD TVS
- ☐ Phase-node copper minimal, outside encoder keep-out on all layers

Sensing

- ☐ Kelvin taps on shunt pads, symmetric, via-free, routed as pairs
- ☐ INA240 REF1=GND / REF2=VDDA; VS=VDDA; 100 nF at each part
- ☐ INA240 outputs → ADC injected channels; verify pin mapping supports simultaneous/paired sampling on ADC1/ADC2 in CubeMX *before* freezing MCU rotation
- ☐ Comparator OCP path to TIM1 BKIN wired; DAC threshold channel confirmed for chosen comparator inputs

Peripherals

- ☐ D+ pairs and D− pairs merged at connector; TVS flow-through; one L4 jog + GND stitch; two separate 5.1 k CC pulldowns
- ☐ Crystal: symmetric, guarded, load caps computed; SWD access present; BOOT0 pulled down; NRST cap
- ☐ CAN split-termination footprint fitted (populate per bus position)
- ☐ AS5047P: VDD-VREG tied (3.3 V mode), 2×100 nF on bottom, SPI mode 1, DIAAGC/AGC check in bring-up firmware plan

Final electrical review

- ☐ Every signal via has a GND via within 1–2 mm (USB, SPI, clocks mandatory)
- ☐ Stitching: perimeter ring, dense field under FETs, every decoupler grounded with its own via
- ☐ No trace crosses a gap between L3 islands on the adjacent L4 layer
- ☐ Startup offset calibration for all three current channels in firmware plan (bridge disabled, store zero codes — do not assume mid-code)

Sources: TI DRV8328 datasheet (SLVSGA6), TI SLVA959B "Best Practices for Board Layout of Motor Drivers", TI INA240 datasheet (SBOS662), ams-OSRAM AS5047P datasheet (DS000324), ST AN2867, ST USBLC6-2 datasheet, JLCPCB capability pages. Verify all pin-level details against the current datasheet revisions before fab.